

Omar Ayman Bakr

AI / LLM Engineer · Machine Learning

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 [LeetCode](#) · 500+ solved

SUMMARY

AI engineer who builds LLM systems and the infrastructure underneath them. Ships production retrieval-augmented services, from vendor-agnostic model and storage abstractions to streamed responses, containerised deployment and monitoring. Trains models as well as serving them, including a hierarchical LSTM for group activity recognition that exceeded the published CVPR 2016 result on the same dataset. **500+ LeetCode problems** solved.

TECHNICAL SKILLS

- **LLM & Retrieval:** RAG, embeddings, vector search (pgvector, Qdrant), semantic chunking, token streaming (SSE), prompt design, LangChain, Ollama, OpenAI / Anthropic / Google / Cohere / NVIDIA NIM APIs
- **Backend:** FastAPI, Pydantic, async I/O, SQLAlchemy 2.0, Alembic, REST APIs, pytest
- **Machine Learning:** PyTorch, Scikit-learn, XGBoost, LightGBM, CatBoost, NumPy, Pandas, Matplotlib, Seaborn
- **Deep Learning:** CNNs (ResNet), LSTMs, Transfer Learning, Focal Loss, Multi-GPU Training, Hydra, TensorBoard
- **Data & Infrastructure:** PostgreSQL, MongoDB, Docker Compose, GitHub Actions (CI/CD), Prometheus, Grafana, Linux, Git, uv
- **Programming:** Python, C++, SQL, C, MATLAB

PROJECTS

[NotebookLLM-minus — Retrieval-Augmented Document Q&A Service](#)

 [Code](#)

LLM / RAG · 2026–Present

- Built a RAG document Q&A service in which no code above the factory layer names a vendor, with **eleven model providers** selectable by configuration or per notebook.
- Abstracted two interchangeable document stores (MongoDB + Qdrant, PostgreSQL + pgvector) behind eight repository interfaces with Alembic migrations.
- Streamed answers over SSE with page-level citations highlighted from word coordinates, cutting extraction on 274 pages from **520s to 79s** across a process pool.
- Ran ingestion as a Celery chain over RabbitMQ, adding a `task_executions` table for per-stage history after Redis results expire and idempotent resubmission on a canonical-args hash.
- Traced weak retrieval to a splitter that orphaned running headers into **110 of 803 chunks** under 50 characters, then merged them into same-page neighbours to double top-5 scores.
- Indexed a `halfvec` cast after finding 2048-dimension embeddings exceeded pgvector's 2000-dimension index ceiling, taking search from **120–185ms to under 1ms** on 24k rows.
- Benchmarked ten Arabic OCR paths (Tesseract on two model packs, EasyOCR, PaddleOCR, a Qwen2-VL fine-tune) on word and character error rate, shipping the winner at **0.172 WER**, 1.15s per page.
- Gated a GitHub Actions pipeline with **594 tests** deploying over SSH to an EC2 host, and tracked ingest duration and time-to-first-token in Prometheus and Grafana.

Keywords: RAG, Vector Search, pgvector, Qdrant, MongoDB, PostgreSQL, Server-Sent Events, Celery, RabbitMQ, Redis, Arabic OCR, Chunking, Repository Pattern, Alembic, GitHub Actions, CI/CD, Docker Compose, Prometheus, AWS EC2

[Volleyball Group Activity Recognition](#)

 [Code](#)

Computer Vision · Deep Learning · 2025–2026

- Reproduced Ibrahim et al., “A Hierarchical Deep Temporal Model for Group Activity Recognition” (CVPR 2016) on the Volleyball Dataset (55 videos, 4,830 clips, 60+GB).
- Built **eight PyTorch baselines** from single-frame ResNet-50 to hierarchical player- and scene-level LSTMs, each isolating one architectural component behind a shared data loader, trainer and evaluator.

- Reached **85.64% accuracy and 0.855 macro-F1** on the best model, above the paper’s published **81.9%**.
- Isolated team-split pooling as the source of **+11.9 accuracy points and +0.154 macro-F1**, resolving the left/right confusion that limited every earlier baseline.
- Ran a control experiment in which a single YOLO26-x classifier on raw frames, using no player annotations, landed within **~7 points** of the full hierarchical model.
- Trained with Hydra configs, AdamW and TensorBoard logging, using `nn.DataParallel` across dual T4 GPUs.

Keywords: Group Activity Recognition, Hierarchical LSTM, ResNet-50, YOLO, PyTorch, Ablation Study, Multi-GPU Training, Hydra, Macro-F1

Credit Card Fraud Detection on Severely Imbalanced Data

Code

Classification · Model Benchmarking · 2025

- Benchmarked **11 models** on 285K transactions at a $< 0.2\%$ fraud rate, implementing Focal Loss from its mathematical definition in PyTorch to test whether an algorithmic fix beats resampling.
- Found XGBoost won at **F1 0.89 and ROC-AUC 0.98**, a 16% F1 gain over the logistic-regression baseline, while focal-loss variants reached only **0.75–0.77**.

Keywords: Imbalanced Classification, Focal Loss, PyTorch, XGBoost, Gradient Boosting, Voting Ensemble, Average Precision, ROC-AUC

New York City Taxi Trip Duration Prediction

Code

Regression · Feature Engineering · 2024

- Engineered **55 features from 10** on **1M+ trips** by integrating five external sources (NYC DOT traffic, weather, borough geometry, holidays) and running a local OSRM routing server in Docker for real road distance and travel time.
- Achieved **RMSE 0.397 and R² 71.6%** with Ridge Regression on 224,734 held-out trips, with train and test RMSE within 0.003 of each other.

Keywords: End-to-End ML Pipeline, Feature Engineering, OSRM / OpenStreetMap Routing, Dockerized OSRM API, Geospatial Analysis, Ridge Regression, Polynomial Features, scikit-learn, Pandas, Docker

EDUCATION

Bachelor of Science, Mechatronics Engineering

2019 – 2024

Helwan University, Cairo, Egypt — Graduated with Honors (Very Good, GPA 3.0/4.0)

Graduation Project: [View Model](#) Built a mathematical model in MATLAB for a green hydrogen generator, controlling input voltage and current while predictively monitoring safety parameters such as operating temperature.

LANGUAGES

Arabic — Native | English — Advanced | German — Basic